

Canonical Ontology

Relational ontology and contract stack

Version 1.0.0-integration.1

Scope

This ontology reconciles the Version 1.0 relational architecture. It is a dependency and evidence architecture, not a universal causal pipeline.

Core commitments

- Interaction is not a primitive substance.
- Opportunity is a typed coordination framework, not a primitive.
- Behavior is attributed under a declared observation model.
- Interpretation, belief, policy, and strategy require explicit inferential treatment.
- Measurement, equivalence, and similarity are contract-relative.

Concepts

Domain commitments

A set of commitments about admissible entities, categories, relations, identity conditions, modalities, and constraints in a domain. In computational work, these commitments may be approximated through ontologies, schemas, contracts, types, or model definitions, but none of those artifacts is automatically a philosophical ontology.

Dependencies: None

Dependents: System state, Semantic assignment

System state

The physical, computational, social, or modeled condition of a declared system at a particular time.

Dependencies: Domain commitments, State transition

Dependents: Projection, Opportunity profile

Semantic assignment

A rule-governed assignment of content, denotation, truth conditions, inferential consequences, or transition behavior to expressions, states, signs, or operations within a specified system and domain.

Dependencies: Domain commitments

Dependents: Semantic state, Projection

Semantic state

The content-bearing assignments and relations currently licensed by a specified semantics for a system or discourse state.

Dependencies: Semantic assignment

Dependents: None

Information

A typed measure or characterization of distinctions, correlations, descriptions, or content relative to a specified source, code, semantic relation, probability model, or use.

Dependencies: None

Dependents: None

Representation

A materially or formally realized vehicle that provides selective access to a declared target through a mapping or correspondence and an interpretive practice.

A representation may preserve, transform, omit, or introduce structure.

Dependencies: Projection

Dependents: Interpretation

Projection

A declared transformation from a target description, system state, or semantic assignment into an accessible representational form.

A projection contract specifies:

- target and authority;
- mapping or construction rule;
- interpretation assumptions;
- preserved properties;
- transformed properties;
- omitted properties;
- introduced properties;
- available operations;
- observer assumptions;
- intended use;
- context and scope;
- uncertainty, tolerance, and failure conditions.

Dependencies: System state, Semantic assignment

Dependents: Representation, Opportunity profile, Scoped invariance

Scoped invariance

A declared property remains invariant across projections when independently specified comparison rules show that it is preserved for a stated target, observer class, context, metric, and tolerance.

Invariance must be typed. Relevant types include:

- ontological;
- semantic;
- informational;
- structural;
- topological;
- computational;
- inferential;
- task;
- behavioral;
- interactional.

Dependencies: Projection, Equivalence

Dependents: None

Interpretation

The application, recovery, construction, negotiation, or revision of semantic relations by an interpreting system under a context, competence, history, and purpose.

Dependencies: Representation, Interaction history

Dependents: Opportunity profile

Interaction participant

Any bounded person, collective, organism, computational process, artificial system, institution, or hybrid arrangement occupying a consequential role within an interaction.

A participant may observe, interpret, act, constrain, authorize, mediate, represent, store, or transform state. Participation does not imply autonomy, intelligence, consciousness, or learning.

Dependencies: Agent

Dependents: Coupling

Agent

An acting interaction participant that selects or produces actions through a policy, rule, control process, intention, or other action-selection organization.

Dependencies: None

Dependents: Interaction participant

Environment

The declared physical, computational, social, symbolic, or mixed system within which action and state transition occur.

Dependencies: None

Dependents: Coupling

Affordance

An action-relevant relation between an agent's effectivities or capabilities and environmental structure that makes a class of action possible. In the Gibsonian sense, perception is not required for the affordance to exist.

Dependencies: None

Dependents: Opportunity profile

Opportunity

A typed and indexed analytical profile coordinating action-availability relations under declared system, agent, projection, task, authority, state, and temporal conditions.

Opportunity is not a new primitive. It coordinates established relations such as feasibility, capability, permission, enablement, reachability, exposure, signaling, belief, expected consequence, reversibility, and uncertainty.

Dependencies: None

Dependents: None

Opportunity profile

For an agent, action, state, projection, and time, an opportunity profile records the relevant typed relations:

- domain-valid;
- effectively capable;
- permitted;
- enabled;
- reachable outcomes;
- exposed;
- signaled;
- believed available;
- expected consequences;
- selected;
- executed;
- successful;
- reversible or recoverable;
- uncertainty and provenance.

Selected, executed, and successful are trace predicates rather than availability sets.

Dependencies: Interpretation, System state, Projection, Affordance, Capability, Permission, Enablement, Reachability, Exposure, Signaling, Believed action

Dependents: Interaction episode

Capability

The effective capacity or resources required for an agent-system coupling to execute an action or bring about an outcome under declared conditions.

Dependencies: None

Dependents: Opportunity profile

Permission

Authorization under a policy, role, rule, authority, or institutional condition for a subject to perform an operation.

Dependencies: None

Dependents: Opportunity profile

Enablement

The condition in which an action's current-state preconditions are satisfied.

Dependencies: None

Dependents: Opportunity profile

Reachability

The existence of at least one admissible transition sequence from a declared initial state to a target state or outcome within specified constraints and horizon.

Dependencies: None

Dependents: Opportunity profile

Exposure

Provision by the current projection of an executable access path to an operation.

Dependencies: None

Dependents: Opportunity profile

Signaling

Perceivable information indicating an operation, condition, consequence, state, or recovery path.

Dependencies: None

Dependents: Opportunity profile

Discoverability

The empirically observed probability and process by which an agent identifies an action or relation under declared task, projection, knowledge, modality, context, and time conditions.

Dependencies: None

Dependents: None

Believed action

An action represented by an agent as available, with an associated confidence and expected consequence.

Dependencies: Interaction history

Dependents: Opportunity profile

Action

A declared intervention, command, operation, or behavior capable of participating in a state transition.

Dependencies: None

Dependents: State transition

State transition

A change from one declared system state to another under a transition relation, process, or event.

Dependencies: Action

Dependents: System state

Observation

A recorded or accessible distinction about state, transition, representation, or outcome available through a specified measurement or observation process.

Dependencies: None

Dependents: Trace

Situated behavior

A bounded, temporally extended trajectory of observable or instrumentally recoverable changes, outputs, or regulated persistence attributed to a system in relation to its environment under a declared observation and attribution model.

Dependencies: None

Dependents: None

Behavioral evidence

A provenance-preserving chain connecting direct observations, system-derived features, latent interpretations, alternative explanations, uncertainty, and falsifying evidence to a bounded research claim.

Dependencies: Measurement contract

Dependents: None

Behavioral equivalence

Equivalence of declared action or outcome distributions for a specified population, task, context, metric, and tolerance.

Dependencies: None

Dependents: None

Interactional equivalence

Equivalence of the available operation-transition-observation structure under specified agent, task, projection, authority, state, context, and tolerance conditions.

Dependencies: None

Dependents: None

Interaction

A family of bounded coupling phenomena in which distinguishable systems or participants become mutually consequential through perturbation, constraint, uptake, response, and recurrence.

Dependencies: Computational Interaction Science

Dependents: None

Interaction episode

A bounded interval in which two or more distinguishable systems are coupled such that perturbations originating in, or mediated by, one system alter the possible or actual state trajectory of another, and the resulting configuration constrains a later trajectory of an originating or co-participating system within a declared boundary and timescale.

Reciprocal closure may be direct, indirect, asymmetric, delayed, probabilistic, or environmentally mediated.

Dependencies: Opportunity profile, Coupling, Token identity, Type identity, Interaction contract

Dependents: None

Coupling

A relation or pathway through which a participant or shared medium can alter another participant's transition structure, reachable states, probabilities, timing, or constraints.

Dependencies: Interaction participant, Environment

Dependents: Interaction episode

Perturbation

A difference propagated through a coupling relation that may change state, transition conditions, probabilities, timing, or reachable outcomes.

Dependencies: None

Dependents: None

Uptake

Incorporation of a perturbation into a participant's state, transition process, control process, or interpreted situation. Uptake need not be semantic or conscious.

Dependencies: None

Dependents: None

Recurrence

Return influence through which a response constrains an originating participant, a co-participant, or the shared coupling configuration.

Dependencies: None

Dependents: None

Interaction contract

A declaration of interaction identity, participants and roles, boundary, timescale, state, coupling, preconditions, events, transitions, protocol, invariants, evidence, failure, recovery, and completion conditions.

Dependencies: None

Dependents: Interaction episode

Interaction morphology

The measurable structural properties of a representation or interaction surface, such as hierarchy, topology, density, alignment, rhythm, symmetry, enclosure, continuity, direction, visual weight, temporal organization, and modality.

Dependencies: None

Dependents: None

Interaction grammar

A typed vocabulary and composition structure for semantic actions and transitions independent of any one input modality or rendering implementation.

Dependencies: None

Dependents: None

Interaction history

The accumulated experience, training, conventions, outcomes, failures, practices, and prior exposure that may change an agent's later beliefs, policies, and interpretation.

Dependencies: None

Dependents: Strategy, Interpretation, Believed action

Trace

A temporally ordered or partially ordered record of observable or attributed events under declared boundaries, granularity, and observation rules.

Dependencies: Observation

Dependents: Measurement contract

Measurement contract

A declaration of the measurand, observation process, operationalization, scale, uncertainty, validity evidence, population, task, projection, comparison conditions, and permitted interpretation.

Dependencies: Trace

Dependents: Interaction similarity, Behavioral evidence

Interaction similarity

A comparison of interaction tokens, types, structures, protocols, opportunities, traces, behaviors, policies, strategies, outcomes, causes, or measurements under an explicit frame, correspondence mapping, admissible transformations, metrics, tolerances, uncertainty, and validity domain.

Dependencies: Measurement contract, Similarity contract

Dependents: Equivalence

Token identity

Historical identity of one bounded interaction episode under declared continuity, participant, boundary, and timescale criteria.

Dependencies: None

Dependents: Interaction episode

Type identity

Membership in the same abstraction-defined interaction class under explicit classification rules.

Dependencies: None

Dependents: Interaction episode

Equivalence

Exact preservation under a declared comparison contract and observable set. Equivalence is always typed.

Dependencies: Interaction similarity

Dependents: Scoped invariance

Similarity

Graded proximity under declared dimensions, representations, alignment rules, metrics, weights, tolerances, and uncertainty.

Dependencies: None

Dependents: None

Incomparability

Failure to establish a defensible common comparison space because boundaries, targets, mappings, scales, observation processes, uncertainty, or validity domains cannot be aligned without unsupported assumptions.

Dependencies: None

Dependents: None

Similarity contract

A declaration of comparison target, frame, representations, correspondence mapping, invariants, variants, observables, admissible and prohibited transformations, typed metrics or predicates, tolerances, uncertainty, validity domain, and final typed profile.

Dependencies: None

Dependents: Interaction similarity

Strategy

An inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response that tends to preserve or improve one or more criteria across a declared class of interaction states.

Dependencies: Policy, Interaction history, Strategy contract

Dependents: None

Strategy contract

A declaration of strategy-bearing system, domain, timescale, criteria, state and observation model, state abstraction, opportunity assumptions, action repertoire, selection policy, information policy, commitments, contingency and recovery rules, switching and stopping conditions, resources, authority, identity invariants, rival strategies, uncertainty, and validity domain.

Dependencies: None

Dependents: Strategy

Policy

A conditional relation from represented states, observations, or beliefs to action distributions.

Dependencies: None

Dependents: Strategy

Fundamental

A representation-independent computational substrate for declaring domain commitments, evolving system state, assigning operational semantics, and producing governed projections whose preservation, transformation, omission, introduced structure, operations, snapshots, and replay can be inspected.

Dependencies: None

Dependents: None

Computational Interaction Science

A research program that studies interaction as an explanatory target by separating and comparing declared domain commitments, state, semantics, projection, representation, interpretation, opportunity, coupling, behavior, measurement, similarity, strategy, and evidence.

CIS must justify itself through comparative explanatory gain, empirical validity, reproducible methods, and error-sensitive predictions. It does not replace HCI, psychology, anthropology, cognitive science, design, computer science, or related disciplines.

Dependencies: None

Dependents: Interaction

Canonical relations

Source	Relation	Target	Authority
Domain commitments	constrains	System state	Canonical ontology/glossary
Domain commitments	constrains	Semantic assignment	Canonical ontology/glossary
Semantic assignment	licenses	Semantic state	Canonical ontology/glossary
System state	source for	Projection	Canonical ontology/glossary
Semantic assignment	source for	Projection	Canonical ontology/glossary

Projection	realizes	Representation	Canonical ontology/glossary
Representation	supports	Interpretation	Canonical ontology/glossary
Interpretation	informs	Opportunity profile	Canonical ontology/glossary
System state	constrains	Opportunity profile	Canonical ontology/glossary
Projection	exposes	Opportunity profile	Canonical ontology/glossary
Opportunity profile	conditions	Interaction episode	Canonical ontology/glossary
Coupling	constitutes	Interaction episode	Canonical ontology/glossary
Interaction participant	participates in	Coupling	Canonical ontology/glossary
Observation	records	Trace	Canonical ontology/glossary
Trace	input to	Measurement contract	Canonical ontology/glossary
Measurement contract	supports	Interaction similarity	Canonical ontology/glossary
Interaction similarity	types	Equivalence	Canonical ontology/glossary
Policy	component of	Strategy	Canonical ontology/glossary
Interaction history	updates	Strategy	Canonical ontology/glossary
Interaction history	conditions	Interpretation	Canonical ontology/glossary
Interaction history	conditions	Believed action	Canonical ontology/glossary
Affordance	contributes	Opportunity profile	Canonical ontology/glossary
Capability	component	Opportunity profile	Canonical ontology/glossary
Permission	component	Opportunity profile	Canonical ontology/glossary
Enablement	component	Opportunity profile	Canonical ontology/glossary
Reachability	component	Opportunity profile	Canonical ontology/glossary
Exposure	component	Opportunity profile	Canonical ontology/glossary
Signaling	component	Opportunity profile	Canonical ontology/glossary
Believed action	component	Opportunity profile	Canonical ontology/glossary
Action	may produce	State transition	Canonical ontology/glossary
State transition	changes	System state	Canonical ontology/glossary
Agent	subtype of	Interaction participant	Canonical ontology/glossary
Environment	mediates	Coupling	Canonical ontology/glossary
Token identity	identifies	Interaction episode	Canonical ontology/glossary
Type identity	classifies	Interaction episode	Canonical ontology/glossary
Similarity contract	specifies	Interaction similarity	Canonical ontology/glossary
Strategy contract	specifies	Strategy	Canonical ontology/glossary
Interaction contract	specifies	Interaction episode	Canonical ontology/glossary
Measurement contract	governs	Behavioral evidence	Canonical ontology/glossary
Projection	tested across	Scoped invariance	Canonical ontology/glossary
Equivalence	expresses	Scoped invariance	Canonical ontology/glossary
Computational Interaction Science	studies	Interaction	Canonical ontology/glossary

Contract stack

The normative contract wording remains in the consolidated contracts source. The stack includes Projection, Opportunity, Interaction, Behavioral Evidence, Measurement, Similarity, Transfer, Strategy, and Prediction contracts.

Appendix A - Candidate Version 2.0 Changes

- Consider whether “Computational Interaction Science” remains the preferred program name after empirical validation; current evidence supports interdisciplinary synthesis rather than disciplinary autonomy.
- Consider a formally ratified distinction between broad ordinary-language interaction and the strict interaction-episode research definition.
- Evaluate whether transfer should become a first-class ontology node in a future normative revision; it is presently represented primarily through contracts and papers.